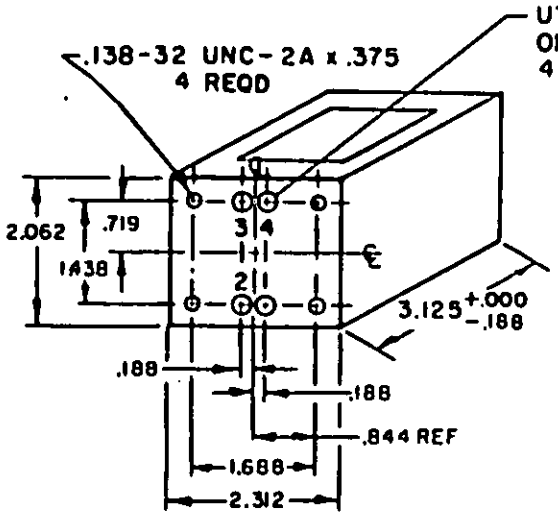
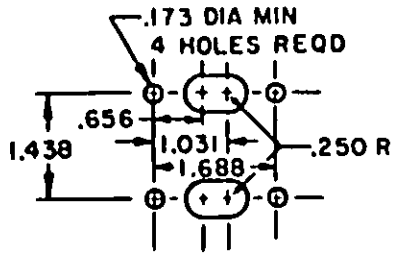
MILITARY SPECIFICATION SHEET

TRANSFORMERS AND INDUCTORS
(AUDIO, POWER AND HIGH-POWER PULSE)
INDUCTOR, POWER, TF4RO4FA

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the inductor described herein shall consist of this document and the latest issue of Specification MIL-T-27.





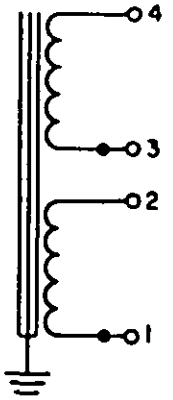
MINIMUM CHASSIS CUTOUT

INCHES	MM
.173	4.39
.188	4.78
.250	6.35
.375	9.52
.656	16.66
.719	18.26
.844	21.44
1.031	26.19
1.438	36.53
1.688	42.88
2.062	52.37
2.312	58.72
3.125	79.38

SERIES-16H
250 V RMS
.080 AMP DC
645 OHMS

120 Hz

PARALLEL-4H
125 V RMS
.060 AMP DC
160 OHMS



WORKING VOLTAGE (PEAK): 1000 V

CIRCUIT DIAGRAM AND MARKING

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerance on case dimensions is +.000 (.00 mm), -.125 (3.18 mm).
4. Tolerance on terminal positioning dimensions is ±.125 (3.18 mm). Terminals fit within minimum chassis cutout.
5. Marking shall be on the side of the case.

FIGURE 1. Dimensions and configuration.

FSC 5950

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REQUIREMENTS: (When numbers in parentheses, i.e., (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings:
- Inductance (1-4):
  - Series connected: 16H minimum.
  - Parallel connected: 4H minimum.
- Voltage (1-4):
  - Series connected: 250 volts rms.
  - Parallel connected: 125 volts rms.
- Current (1-4):
  - Series connected: 0.080 ampere dc.
  - Parallel connected: 0.160 ampere dc.
- DC resistance:
  - (1-2): 324.5 ohms maximum.
  - (3-4): 385 ohms maximum.
- Working voltage (peak): 1,000 volts.
- Power: 2.08 watts maximum (each winding).

Design and construction:
- Dimensions and configuration: See figure 1.
- Duty cycle: Continuous.
- Case: Metal encased.
  - Material: Steel.
- Weight: 2 pounds maximum.
- Terminals: Solder terminal, United Transformer Co. NC-3 or equivalent.
  - Height: 0.469 inch, maximum.
- Operating temperature range: -55°C to +105°C.

Terminal strength: MIL-STD-202, method 211, test condition A, 5 pounds and test condition E, 8 ounce-inches.

Dielectric withstanding voltage (at sea level): 2,400 volts rms.

Electrical characteristics:
- Inductance:
  - Series connected: With 250 volts rms, 120 Hz, and 80 mA dc applied to (1-4), the inductance shall be 16 Henries, minimum.
  - Parallel connected: With 125 volts rms, 120 Hz, and 160 mA dc applied to (1-4), the inductance *shall be 4 Henries, minimum.*
- DC resistance:
  - (1-2): 324.5 ohms, maximum.
  - (3-4): 385 ohms, maximum.

Temperature rise: 40°C with 250 volts rms, 120 Hz, and 80 mAdc applied to (1-4) with (2-3) connected at an ambient temperature of 65°C.

Shock (specified pulse): MIL-STD-202, method 213, test condition H.

Marking location: See figure 1.

Part number: M27/209-01.

Note: MIL-T-27/209 supersedes MS90009 which is inactive for design after the date of this specification sheet. M27/209-01 supersedes MS90009-2.

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Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - SH, OS
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AR, WC, ME
Navy - MC, AS
Air Force - 19

Preparing activity:
Army - EL

Agent:
DLA - ES

(Project 5950-0538-71)

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